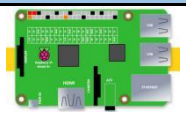
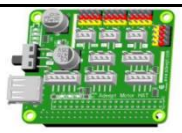
# Lesson 4 How to Control onboard LED

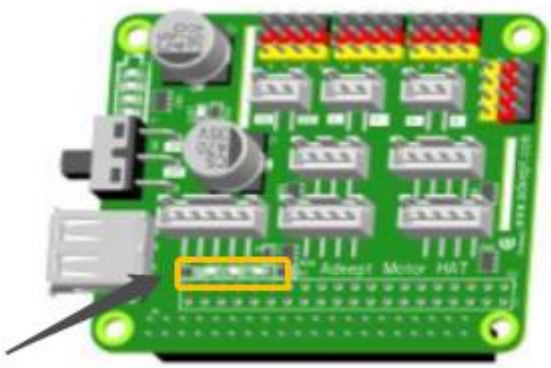
In this lesson, we will learn how to control onboard LED.

## 4.1 Components used in this course

| Components | Quantity | Picture |
|---|---|---|
| Raspberry Pi | 1 |  |
| Motor HAT | 1 |  |

## 4.2 Wiring diagram (Circuit diagram)

Use LED0, LED1, LED2, the pin pairs used are GPIO5,GPIO6,GPIO13:



## 4.3 Run the code

1. Remotely log in to the Raspberry Pi terminal.

```
Linux raspberrypi 4.19.118-v7l+ #1311 SMP Mon Apr 27 14:26:42 BST 2020 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Aug 29 08:17:49 2020 from 192.168.3.208

SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set
 a new password.

pi@raspberrypi:~ $
```

2. Enter the command and press Enter to enter the folder where the program is located:

```
cd adeept_picar-b/server/
```

```
pi@raspberrypi:~ $ cd adeept_picar-b/server/
pi@raspberrypi:~/adeept_picar-b/server $
```

3. View the contents of the current directory file:

**ls**

```
pi@raspberrypi:~/adeept_picar-b/server $ ls
app.py              flask-video-streaming  Instruction.txt    __pycache__        setup.py
appserverAP.py      FPV.py                 Kalman_filter.py   raspi-config.py    speech.py
appserver.py        FPVtest.py             LEDapp.py          RGB.py             SR.py
base_camera.py      functions.py           LED.py             robotLight.py      switch.py
camera_opencv.py    GUIfindline.py         move.py            RPIservo.py        ultra.py
config.txt          GUImove.py             OLED.py            server.py          webServer.py
dist                image2.jpg             onboardLED.py      serverTest.py
findline.py         info.py                PID.py             servo.py
pi@raspberrypi:~/adeept_picar-b/server $
```

4. Enter the command and press Enter to run the program:

**sudo killall python3**

```
pi@raspberrypi:~/adeept_picar-b/server $
pi@raspberrypi:~/adeept_picar-b/server $ sudo killall python3
```

If an error is reported when running the LED.py program, please check the Q**&A** of this tutorial.

**sudo python3 onboardLED.py**

```
pi@raspberrypi:~/adeept_picar-b/server $ sudo python3 onboardLED.py
```

5. After successfully running the program, you will see the on-board leds flashing in sequence.

6. When you want to terminate the running program, you can press the shortcut key "Ctrl + C" on the keyboard.

## 4.4 Main code program

For the complete code, please refer to the file LED.py.

```
import time
from rpi_ws281x import *
```

Import dependency.

```
pin_led_1 = 5
pin_led_2 = 6
pin_led_3 = 13
```

```
GPIO.setmode(GPIO.BCM)
GPIO.setup(pin_led_1, GPIO.OUT)
GPIO.setup(pin_led_2, GPIO.OUT)
GPIO.setup(pin_led_3, GPIO.OUT)


def control_led(pin, state):
    GPIO.output(pin, state)
```

Build onboard LED control.

```
try:
    while True:
        control_led(pin_led_1, True)
        time.sleep(0.5)
        control_led(pin_led_1, False)
        time.sleep(0.5)

        control_led(pin_led_2, True)
        time.sleep(0.5)
        control_led(pin_led_2, False)
        time.sleep(0.5)

        control_led(pin_led_3, True)
        time.sleep(0.5)
        control_led(pin_led_3, False)
        time.sleep(0.5)

except KeyboardInterrupt:
    pass
```